

Zero Trust, Zero Hassle: Simplifying Secure Internet Access

Please scan the QR code to check in this session

Pim Jacobs

Technology Lead Security @ InSpark
& Microsoft Security MVP

- Focus on the full Microsoft Entra portfolio
- One of the founders of the **Dutch Microsoft Entra Community**, join our meetup!



- Blog: identity-man.eu
- Skiing | Soccer | F1 | Time with the kids



in



X



Show of hands

- Who in this room is using Entra Internet Access or has configured it in production?
- Who in this room is blocking Gen AI access to end users?
- Who is familiar with Defender for Threat Intelligence?

The internet threat landscape

Attacks have become more complex & sophisticated



Unaddressed internet
attack paths

84%

of attack paths involve
internet exposure



Excessive access
permissions

>80%

of organizations have users
with excessive access
permissions



Increase in
cybercrime

7,000+

password attacks
per second in 2024



Increase in
AI risks

57%

of security leaders report an
increase in security incidents
due to AI usage

The rise of AI in the workplace has been staggering



75%

Seventy-five percent of knowledge workers already use AI at work, a figure that has doubled in just six months.



1 of 4

Yet only one in four enterprise AI initiatives is secured.



93%

93 percent of companies acknowledge the risks but only nine percent feel prepared to address them.



78%

of employees bringing their own AI tools into the workplace.

Legacy network security stacks are no longer sufficient

The rise of cloud workloads, AI, and hybrid work is straining traditional corporate networks and network security models, resulting in security risks and poor user experience.

Challenges with conventional approaches

1

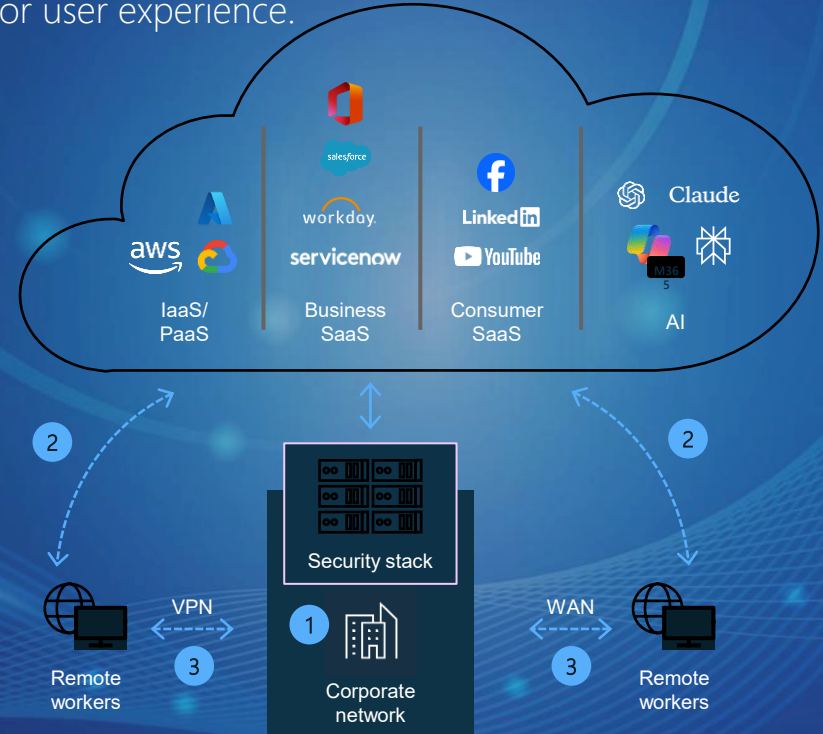
Dramatic traffic increase strains network capacity and on-prem security stack. Sub optimal user experience on account of traffic hair-pinning.

2

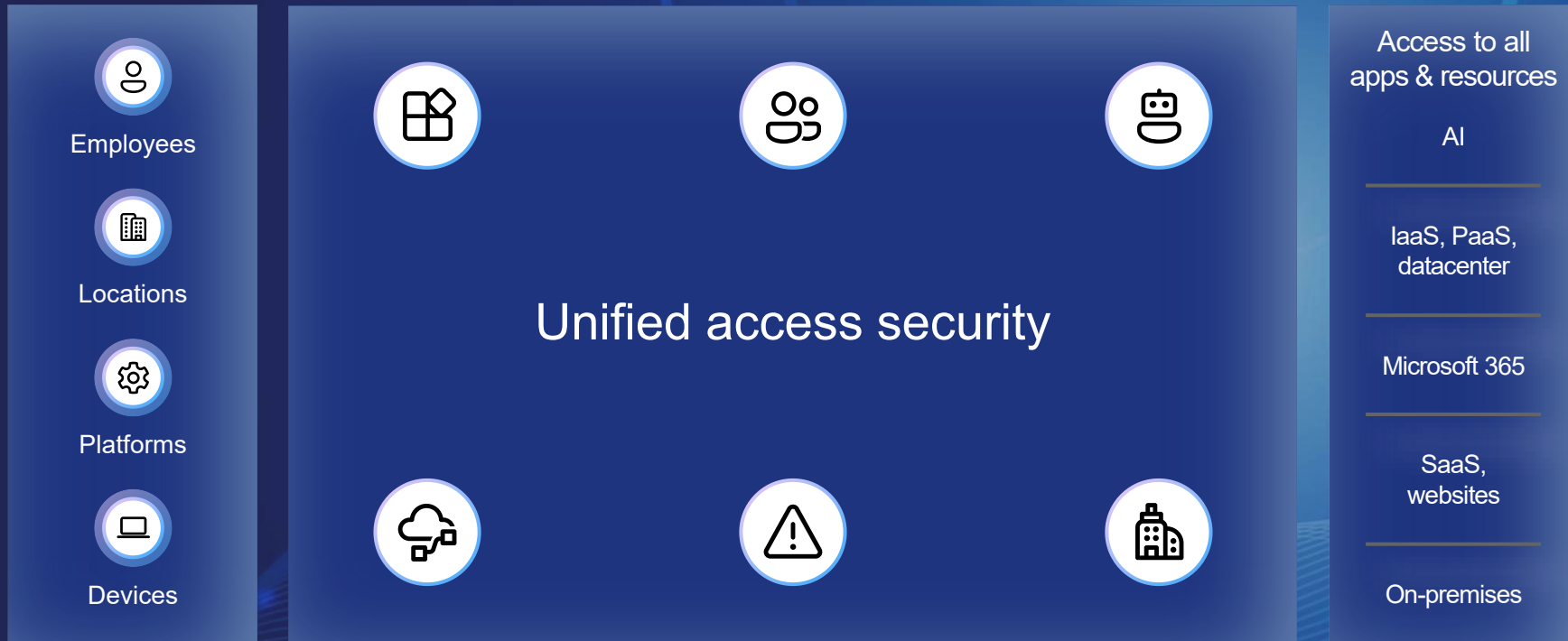
Users circumvent IT controls and access resources directly.

3

Compromised users/devices can move laterally on traditional corporate networks.

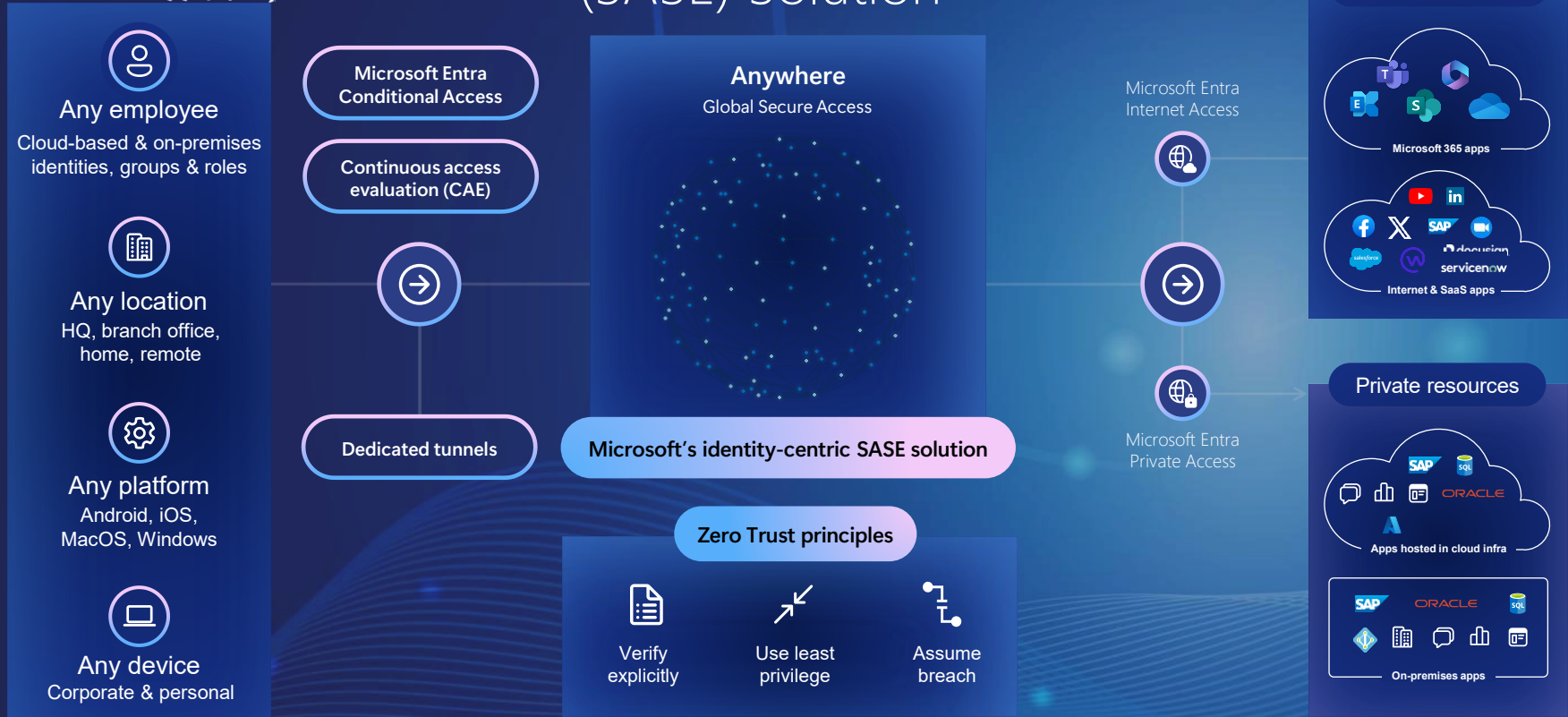


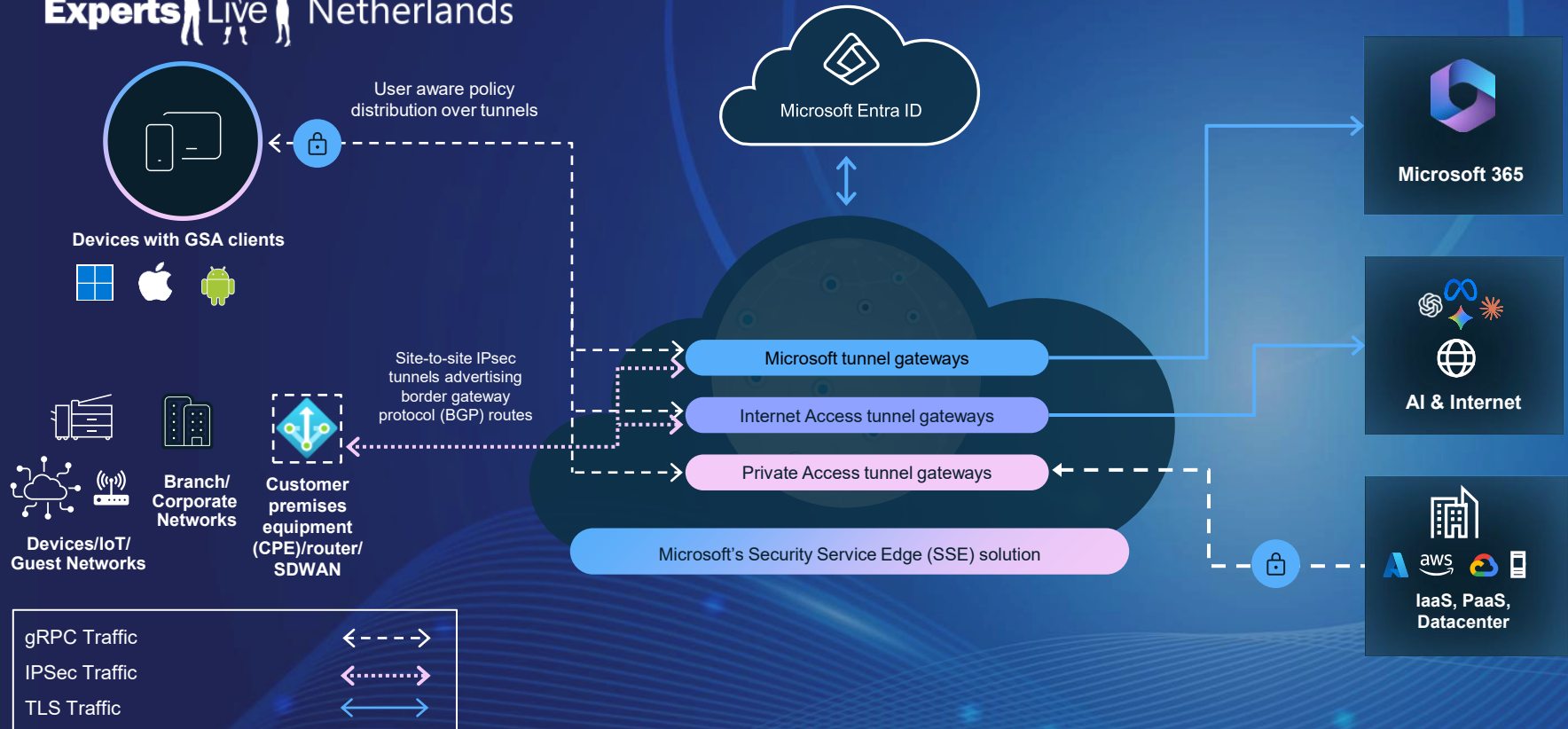
Addressing evolving threats requires a new approach



Global Secure Access

Microsoft's Secure Access Service Edge (SASE) Solution





Microsoft's SASE Solution Unique Differentiators

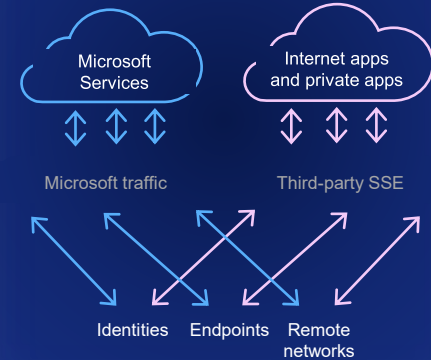
Unified approach across identity and network access



Microsoft's global private wide area network



Flexible deployment options



Secure access to any app or resource, from anywhere, with an identity-centric SASE solution

Enforce Conditional Access controls

Extend Conditional Access and Continuous Access Evaluation (CAE) to any application or resource



Streamline network security

Escape the complexity and cost of traditional stand-alone network security tools with comprehensive cloud-delivered services



Deliver a fast, consistent hybrid work experience

Boost hybrid work productivity with fast and seamless access through a globally distributed secure network edge



Entra Internet Access



Extend Conditional Access
to web and AI traffic



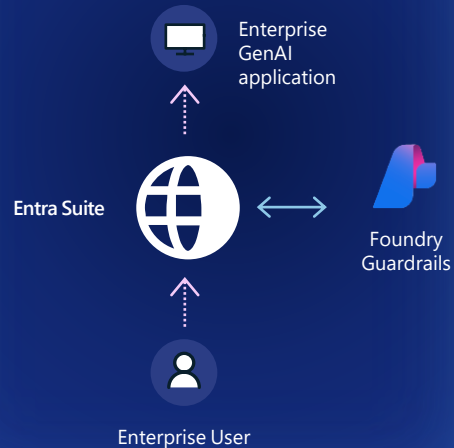
Defend users from internet
threats



Extend universal run-time
protection to any AI

Prompt Injection Protection

Protect enterprise GenAI apps from XPIA/UIPA attacks, protect jailbreak



Network-based Data Security

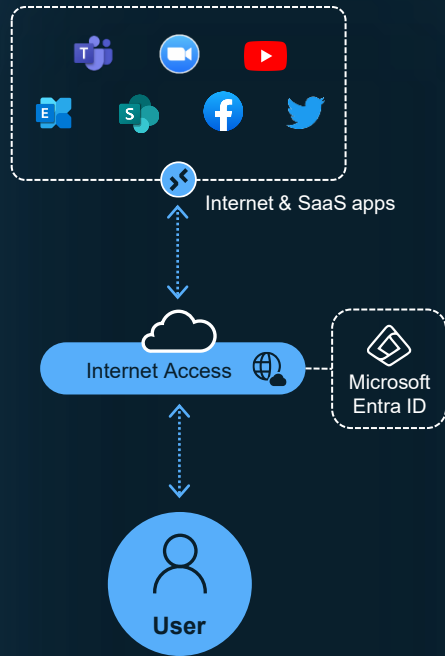
Prevent corporate data exfiltration using network file filtering



Discover and onboard shadow AI

Discover apps, agents, data source, etc. being used and seamless onboard and attribute them to include in Conditional Access policies





Key scenarios and use cases

Stop prompt injection attacks

Protect sensitive data with network file filtering

Secure agents in AI gateway controls

Block malicious destinations with threat intelligence filtering

Prevent access to risky web content

Inspect encrypted traffic with transport layer security (TLS) inspection

Protect against token replay attacks

Client Deployment



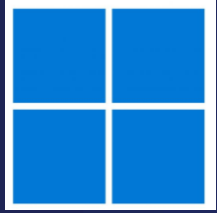
Microsoft



SOFTWARE
CENTRAL



Client OS Support



Windows 10/11

- Windows 11 Arm64 only
- No multi-session VDI support & VM's running on host with GSA client



MacOS 13 or newer

- Device registered in Microsoft Entra
- Use the Microsoft Enterprise SSO plug-in

iOS

- iOS 15.0 and above
- iPadOS supported as well



Android

- Android 10.0 and above
- Android Go not supported

Windows Client policies

Select the PowerShell scripts to expand them.

▼ PowerShell detection script

HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa\Kerberos\Parameters

Note

If the `Parameters` key isn't listed under `Kerberos`, you must create it.

Modify the following `FarKdcTimeout` registry entry

- Entry: `FarKdcTimeout`
- Type: `REG_DWORD`
- Default value: `0 (minutes)`

```
"RestrictNonPrivilegedUsers" = 0
"Type": "REG_DWORD"
}
```

TLS Inspection certificate

Disable QUIC in browsers

Disable DNS-over-HTTPS in browsers

Pro-active remediations for tray icon settings

FarKdcTimeout registry key (PA Only)

System Extensions

Custom Configuration Profile

Custom configuration profile name ⓘ

com.microsoft.globalsecureaccess.applicationproxy

Configuration profile file ⓘ

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.
3 <plist version="1.0">
4   <dict>
5     <key>PayloadUUID</key>
6     <string>87cbb424-6af7-4748-9d43-f1c5dc
7     <key>PayloadType</key>
8     <string>Configuration</string>
9     <key>PayloadOrganization</key>
10    <string>Microsoft Corporation</string>
11    <key>PayloadIdentifier</key>
12    <string>com.microsoft.globalsecureacce
13    <key>PayloadDisplayName</key>
14    <string>Global Secure Access Proxy Cor
15    <key>PayloadDescription</key>
```

TLS Inspection certificate

Allow system extensions for Global Secure Access

Allow transparent application proxy

Custom policy for tray icon settings

Android Client policies



basics

Name D-APPCONF-InSpark-AND-ManagedDevices-GSA

Description No Description

Device enrollment type Managed devices

Platform Android Enterprise

Profile Type Fully Managed, Dedicated, and Corporate-Owned Work Profile Only

Targeted app Microsoft Defender: Antivirus

Settings

Permissions

Not configured

Configuration Settings

Configuration key	Value type	Configuration value
Global Secure Access	integer	3
GlobalSecureAccessPrivateChannel	integer	3



Work securely from anywhere

Global Secure Access lets you connect safely to your organization's resources and private apps. [Learn more.](#)

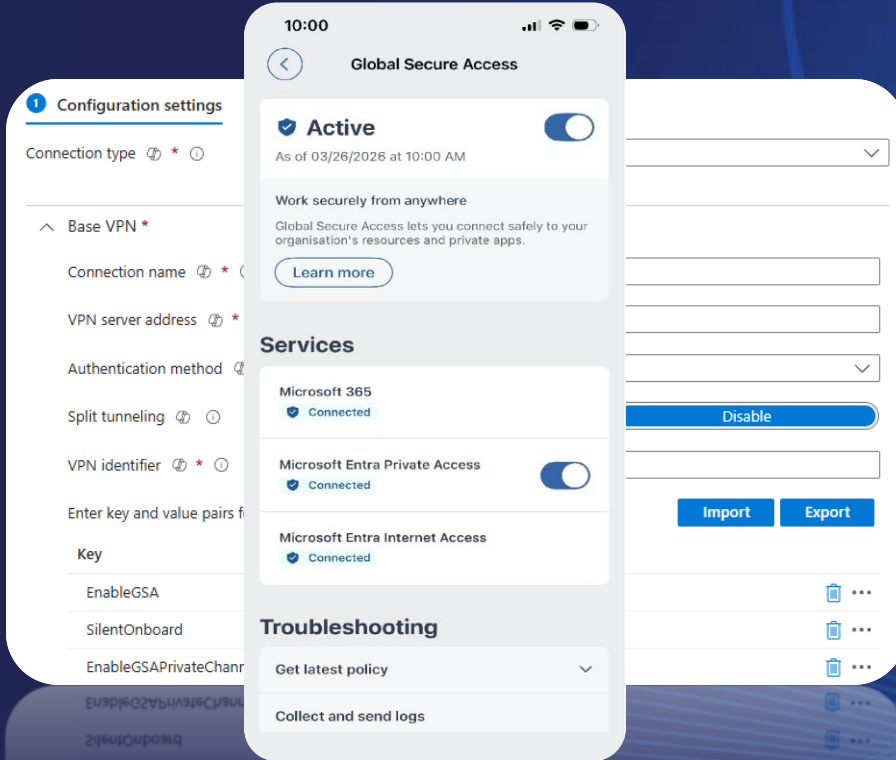
TLS Inspection certificate

App protection policies deployed for defender client

Both Internet Access and Private Access are separate settings

Defender for Endpoint deployed

iOS Client policies



TLS Inspection certificate

VPN policy deployed for defender client

Both Internet Access and Private Access are separate settings

Defender for Endpoint deployed

Configuring Entra Internet Access

Enable Traffic Forwarding Profile for M365 and Regular Internet Access

Enable Conditional Access Signaling

Create TLS Inspection Certificate & TLS Inspection policies

Create Web content filtering policies

Create Threat Intelligence policies

Create Content Filtering & Prompt Injection Policies

Create Security Profiles and CA Policies



Microsoft Entra admin center

Search resources, services, and docs (G+)

Copilot

pim.jacobs@weareinspa...
INSPARK B.V. (INSPARK.NL)

Home

Security Copilot agents

Copilot chat

Favorites

Entra ID

ID Protection

ID Governance

Verified ID

Global Secure Access

What's new

Billing

Security Store

Diagnose & solve problems

New support request

Microsoft Entra

Give me a summary of active high-priority identity recommendations.

Show me the count of risky users, risky service principals, and stale apps.

List my top identity governance action items.

The Conditional Access Agent identified 4 suggestions

Close policy gaps to instantly elevate your zero trust posture. [Learn more](#)

View suggestions

InSpark B.V.

Tenant ID 2b55fa1c-0add-4120-a8e4-99... [🔗](#)

Primary domain inspark.nl [🔗](#)

1,873

View users

1,642

View groups

793

View devices

1,028

View apps

PJ Pim Jacobs

296acf05-ba37-462f-a4bb-f023380266be [🔗](#)

[View user profile](#)

My role assignments

3

High privileged role assignments

Other role assignments

[Manage my roles](#)

Users at high risk

1 user detection with risk level "high" in the last 3 months. [Learn more](#)

No change in the last 30 days

[View high risk users](#)

Shortcuts

Add

User sign-ins

Audit logs

Authentication Methods

Blocked users

Domain names

Unused service principals

Manage tenants

Named locations

Cross-tenant Access Policies

Tenant restrictions

Risk Based Conditional Access policies

Risky sign-ins

Lifecycle workflows

Get the most out of your licenses and subscriptions

Deployment guides

Use these deployment guides to plan, configure and deploy Microsoft Entra capabilities

Suggestions

Modernize remote access to on-premises apps with MFA for each app [🔗](#)

Microsoft Entra plan

Entra Suite

Tenant status

Identity Secure Score 95.22%

[View recommendations](#)

Microsoft

SOFTWARE CENTRAL

Capgemini

inforcer

DELL Technologies

nerdio

SUPER VISION

INTERSTELLAR

kpn Partner Network

INSPIRE

New InPrivate

+ New tab Ctrl+T



InPrivate browsing

InPrivate search with Microsoft Bing



✓ What InPrivate browsing does

Deletes your browsing info when you close all InPrivate windows
Saves collections, favorites, and downloads (but not download history)
Prevents Microsoft Bing searches from being associated with you

✗ What InPrivate browsing doesn't do

Hide your browsing from your school, employer, or internet service provider
Give you additional protection from tracking by default
Add additional protection to what's available in normal browsing

Always use "Strict" tracking prevention when browsing InPrivate



If this is off, we'll use the same tracking prevention setting as a normal browsing window

Cookies aren't shared during InPrivate

InPrivate browsing keeps cookies private to help prevent sites from sharing your data across the web. If a site doesn't work, you can temporarily allow cookie access.

[↓ More details](#)



Entra Internet Access Results

-  TLS Inspection enabled
-  Web categories blocked
-  Malicious website blocked by threat intelligence
-  Specific files blocked for upload to websites categorized with AI
-  Malicious prompts are being blocked
-  Custom block page for blocked websites for GSA
-  Traffic logs can be used for troubleshooting & Insights & Analytics for AI usage

M365 Traffic Profile & CA

=

Phishing resistency for M365



Current Mailbox

kondor wessels

File

Home

Send / Receive

View

Help

Search

Try the new Outlook

From

Sent To

Subject

Has Attachments

Unread

Categorised

Flagged

Important

Include Deleted Items

Close Search

Favourites

Sent Items

Inbox - pim.jacobs@inspark.nl69

Deleted Items27

Inbox - pim.jacobs@hotmail.com9

Inbox - pim.jacobs@jacobsa...44

Onbelangrijke e-mail

Junk Email[6]

Outbox

PST Import

Quick Step Settings

RSS Feeds

RSS-abonnementen

Synchronisatieproblemen

Troep

Search Folders

Groups

Info

Inbox13

Drafts

Sent Items

Deleted Items

Archive

Conversation History

Junk Email[13]

Outbox

Results

By Date

From

Subject

Received

Size

Count

Mention

Top Results

Barry Oudejans

Kondor Wessels project documentation review

Fri 8/23/2024 11:16 AM

8...

Michiel Konijn

Preview - Learn & lunch: Veilige gebruikerstoegang met de Microsoft Entra Suite

Wed 8/21/2024 5:11 PM

6...

Older

Pim Jacobs

Advanced Specializations - Security

Fri 2/2/2024 4:40 PM

5...

Barry Oudejans; Jelmer Vorstenburg

Microsoft CIAM solution for Keesing Media Group

Fri 9/29/2023 7:55 AM

1 ...

Search Complete.

Items: 6

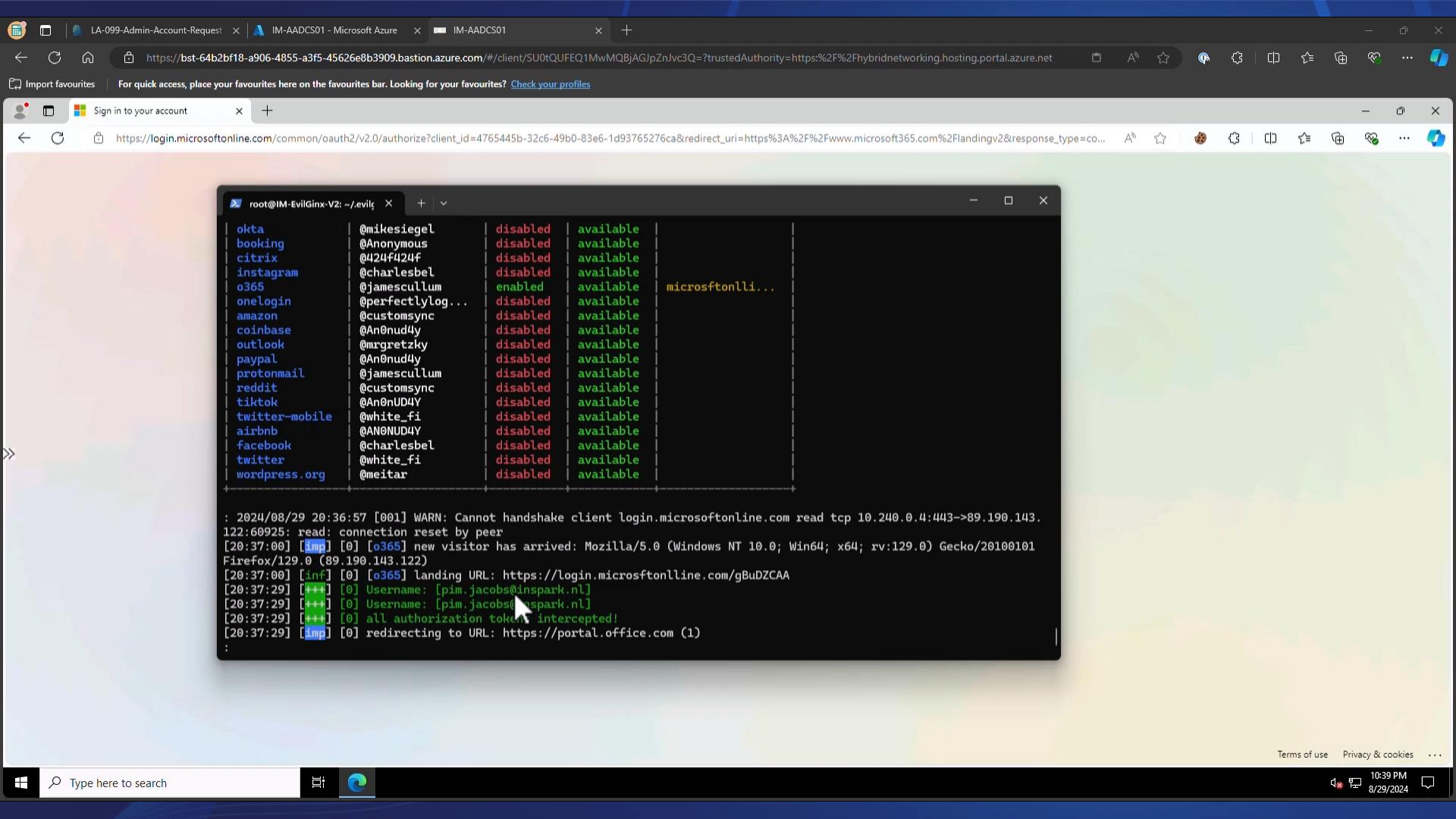
Reminders: 2

All folders are up to date.

Connected to: Microsoft Exchange

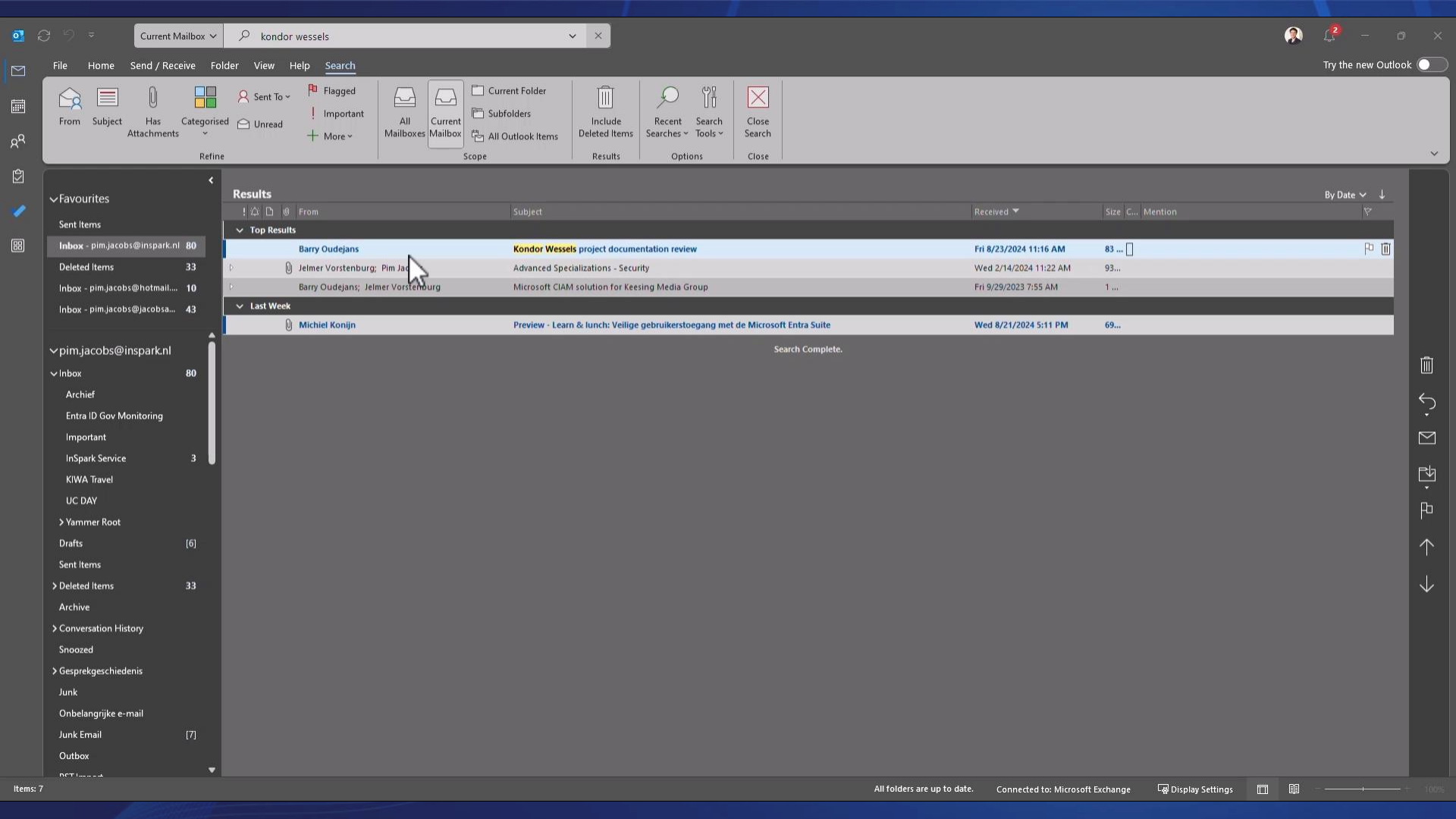
Display Settings

100%



okta	@mikesiegel	disabled	available	
booking	@Anonymous	disabled	available	
citrix	@424f424f	disabled	available	
instagram	@charlesbel	disabled	available	
o365	@jamescullum	enabled	available	microsfonlli...
onelogin	@perfectlylog...	disabled	available	
amazon	@customsync	disabled	available	
coinbase	@An0nUD4y	disabled	available	
outlook	@mrgretzky	disabled	available	
paypal	@An0nUD4y	disabled	available	
protonmail	@jamescullum	disabled	available	
reddit	@customsync	disabled	available	
tiktok	@An0nUD4Y	disabled	available	
twitter-mobile	@white_fi	disabled	available	
airbnb	@AN0NUD4Y	disabled	available	
facebook	@charlesbel	disabled	available	
twitter	@white_fi	disabled	available	
wordpress.org	@meitar	disabled	available	

```
: 2024/08/29 20:36:57 [001] WARN: Cannot handshake client login.microsoftonline.com read tcp 10.240.0.4:443->89.190.143.
122:60925: read: connection reset by peer
[20:37:00] [imp] [0] [o365] new visitor has arrived: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:129.0) Gecko/20100101
Firefox/129.0 (89.190.143.122)
[20:37:00] [inf] [0] [o365] landing URL: https://login.microsoftonline.com/gBuDZCAa
[20:37:29] [+++] [0] Username: [pim.jacobs@inspark.nl]
[20:37:29] [+++] [0] Username: [pim.jacobs@inspark.nl]
[20:37:29] [+++] [0] all authorization tokens intercepted!
[20:37:29] [imp] [0] redirecting to URL: https://portal.office.com (1)
:
```



Conclusion

Client configuration is crucial for success

For clientless scenario's (VDI) explicit forward proxy can be used

TLS Inspection is mandatory for optimal results

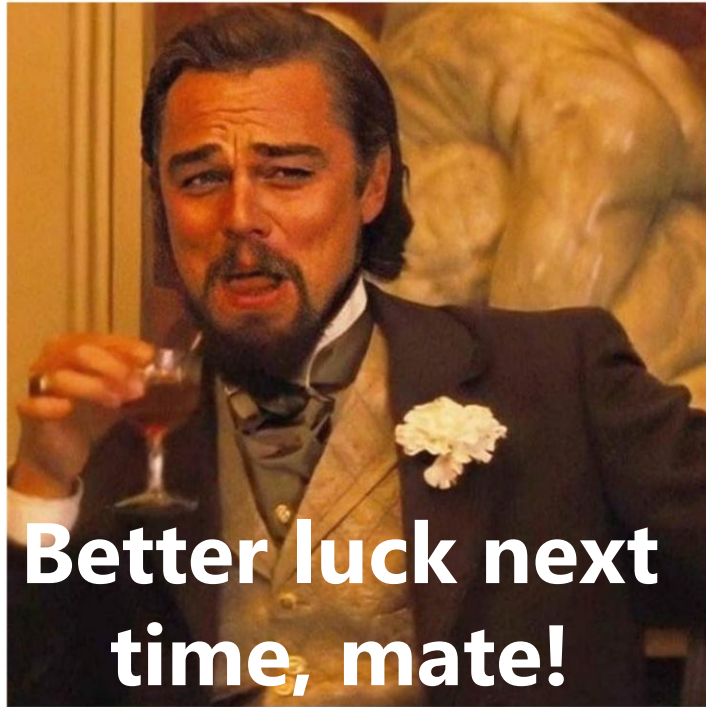
Create separate CA policies per functionality for easy exclusion

Start small, grow big

Microsoft Purview integration available for file content scanning



Next steps on M365 Internet Access



Deploy the GSA client & basic tenant configuration (P1)

Enable the M365 traffic profile (P1)

Enable CA Signaling

Create the compliant network check CA policy (P1)

Enable the Internet Access Traffic profile

Start with TLS Inspection

And expand further from this point forward.

Integrate with SOC operations

Companies not
protecting user's
internet access

Companies
protecting user's
internet access





Please evaluate this session by scanning the QR code!

THANK YOU

Are there any questions?

Or drop by the Ask the Experts

booth Loop after this session!





Next session 14:00 – 14:50

**Every Device Is a Security Boundary: Secure It
by Design – Morten Knudsen**

